METHOD AND SYSTEM FOR DE-ANONYMIZING RADIO FREQUENCY SIGNALS VIA A DISTRIBUTED SENSOR ARRAY

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Title: METHOD AND SYSTEM FOR DE-ANONYMIZING RADIO FREQUENCY SIGNALS VIA A DISTRIBUTED SENSOR ARRAY

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Abstract

A system and method for identifying the source of a radio frequency (RF) signal, such as a cellular communication, by using a distributed array of passive RF sensors. The system comprises a plurality of sensors, which may be embodied in vehicles, arranged in a predetermined geometric pattern around an RF source, such as a cellular tower. This arrangement ensures any given signal is received by at least two sensors. A central processing unit calculates the signal's precise direction of origin by analyzing the differences in the signal's Time of Arrival (ToA) at each sensor. This directional vector is then correlated with event logs from the RF source's operator (e.g., a telecommunications company), which contain precise timestamps for communication events. By matching the signal's time and direction, the system can de-anonymize the communication and identify the specific user, even if the signal content is encrypted. A further aspect of the invention involves a dual-use deployment for

counter-intelligence purposes.

Background of the Invention

Identifying the source of radio frequency (RF) signals is a fundamental challenge in signals intelligence (SIGINT) and law enforcement. While the content of communications is often protected by encryption, the metadata—such as the identity and location of the communicating parties—is highly valuable. Traditional methods for locating an RF source often rely on triangulation from multiple fixed towers, which can provide a general location but may lack the precision to identify a specific user among many in a small area. Furthermore, access to the communication content or unencrypted metadata is often restricted. There is a need for a method that can precisely identify the originator of a specific communication event using only the signal itself and commonly available external data, without needing to decrypt the signal's content.

Summary of the Invention

The present invention provides a system and method for de-anonymizing RF signals by correlating high-precision directional data with telecommunication event logs. The system uses a plurality of RF sensors, which can be discreetly integrated into vehicles, positioned in a specific, scientifically determined geometric arrangement around an RF source, such as a cellular tower. The key to this arrangement is that any signal emitted by a user and received by the tower is guaranteed to also be received by at least two of the sensors in the array.

A central processing unit, communicatively coupled to the sensor array, continuously receives signal data. When a signal of interest is detected, the system records the precise Time of Arrival (ToA) at each sensor that received it. Based on the nanosecond-level differences in ToA between the sensors, the system calculates a highly accurate vector pointing to the signal's origin.

This directional vector is then combined with a second source of data: the event logs from the telecommunications provider that operates the cell tower. These logs contain an entry for every communication event (e.g., a text message sent, a call initiated) with a precise timestamp. The system queries these logs for an event that occurred at the exact millisecond the signal was detected. By correlating the precise time with the precise direction, the system can identify the specific user account responsible for the transmission, as the probability of two different users in the exact same direction initiating a logged event at the exact same millisecond is effectively zero. A further embodiment describes a dual-use deployment model

where sensors are placed in specific configurations for both broad area triangulation and targeted counter-intelligence surveillance of specific premises.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government, its military, and its defense agencies a perpetual, universal, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, national security, and non-commercial purposes, including applications related to supporting the country's allies or the country's interests. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a system diagram illustrating the overall architecture, showing the relationship between the RF source, the distributed sensor array, and the data correlation unit.
FIG. 2 is a conceptual diagram illustrating the dual-use deployment of the sensor array for both wide-area triangulation and specific counter-intelligence monitoring.

Detailed Description of the Invention

Referring to **FIG. 1**, the system for de-anonymizing RF signals is shown. An RF source, such as a cellular tower (101), provides service to multiple users, including a target user (102). A plurality of RF sensors (103), which are preferably concealed within innocuous vehicles, are placed in a predetermined geometric arrangement around the tower (101). When the target user (102) initiates a communication, an RF signal (104) travels to the tower. This same signal is contemporaneously detected by at least two of the sensors (e.g., Sensor A and Sensor B).

The sensors transmit the detected signal data, including the precise Time of Arrival (ToA), to a Central Processing Unit (105). The CPU executes a ToA analysis algorithm (106) that compares the arrival times at Sensor A and Sensor B. Based on the time delta, it computes a precise directional vector (107) pointing back to the origin of the signal (102). The system then accesses the Telecom Event Logs (108), which are provided by the operator of the tower (101). The CPU performs a correlation (109) between the signal's detection

time and the timestamps in the logs. The system identifies a unique log entry (110) that matches both the time and the direction, thereby revealing the user's identity associated with the signal.

Referring to **FIG. 2**, a dual-use deployment strategy is illustrated. In a residential area or operational hub, a first set of sensor vehicles (201) are parked on public streets. These sensors form a wide-area array, used for the general triangulation and direction-finding of signals within the neighborhood, as described in FIG. 1. A second, distinct set of sensor vehicles (202) are parked in the driveways of specific properties (203) known to be associated with a group of interest. This second set of sensors serves a specialized counter-intelligence function. They are positioned to have a high-sensitivity listening posture for any RF signals originating from *within* or immediately around the premises (203). If a covert listening device (a "bug") (204) were planted inside the property by an outside agency and began to transmit, the signal (205) would be immediately detected by the driveway sensors (202). This allows the system to not only monitor external communications but also to detect and flag attempts at technical surveillance against its own assets.

Claims

What is claimed is:

1. A system for identifying the source of a radio frequency (RF) signal, comprising:
    - a plurality of RF sensors physically distributed in a predetermined arrangement around an RF signal receiver;
    - a processing unit communicatively coupled to the plurality of RF sensors;
    - wherein the processing unit is configured to:
        - receive signal data, including a Time of Arrival (ToA), from at least two of said RF sensors that have detected the same RF signal;
        - calculate a directional vector to the origin of the RF signal based on differences in the ToA at said sensors;
        - access an external log of communication events associated with the RF signal receiver, said log containing timestamps and user identifiers; and
        - correlate the directional vector and the time of the signal's detection with an entry in said log to identify a user associated with the RF signal.

2. The system of claim 1, wherein the RF sensors are integrated into vehicles.

3. The system of claim 1, wherein the RF signal receiver is a cellular communications tower.

4. The system of claim 1, wherein the predetermined arrangement is a geometric pattern calculated to ensure a signal from any direction is detected by at least two sensors.

5. A method for de-anonymizing a radio frequency (RF) communication, comprising the steps of:

    - arranging a plurality of RF sensors in a geometric pattern around a central RF receiver;
    - detecting an RF signal at two or more of said RF sensors;
    - determining the Time of Arrival (ToA) of the signal at each of the two or more sensors;
    - calculating a direction of origin for the signal based on the difference in the ToA between the sensors;
    - obtaining a log of time-stamped communication events from the operator of the central RF receiver; and
    - matching the direction of origin and the time of the signal's detection to a specific event in the log to identify the sender of the communication.

6. The method of claim 5, further comprising the step of deploying a first subset of said RF sensors on public thoroughfares for calculating the direction of origin, and deploying a second subset of said RF sensors on private property to detect signals originating from said property.

7. The method of claim 6, wherein the detection of a signal originating from said private property is used to identify a covert surveillance device.

Drawings

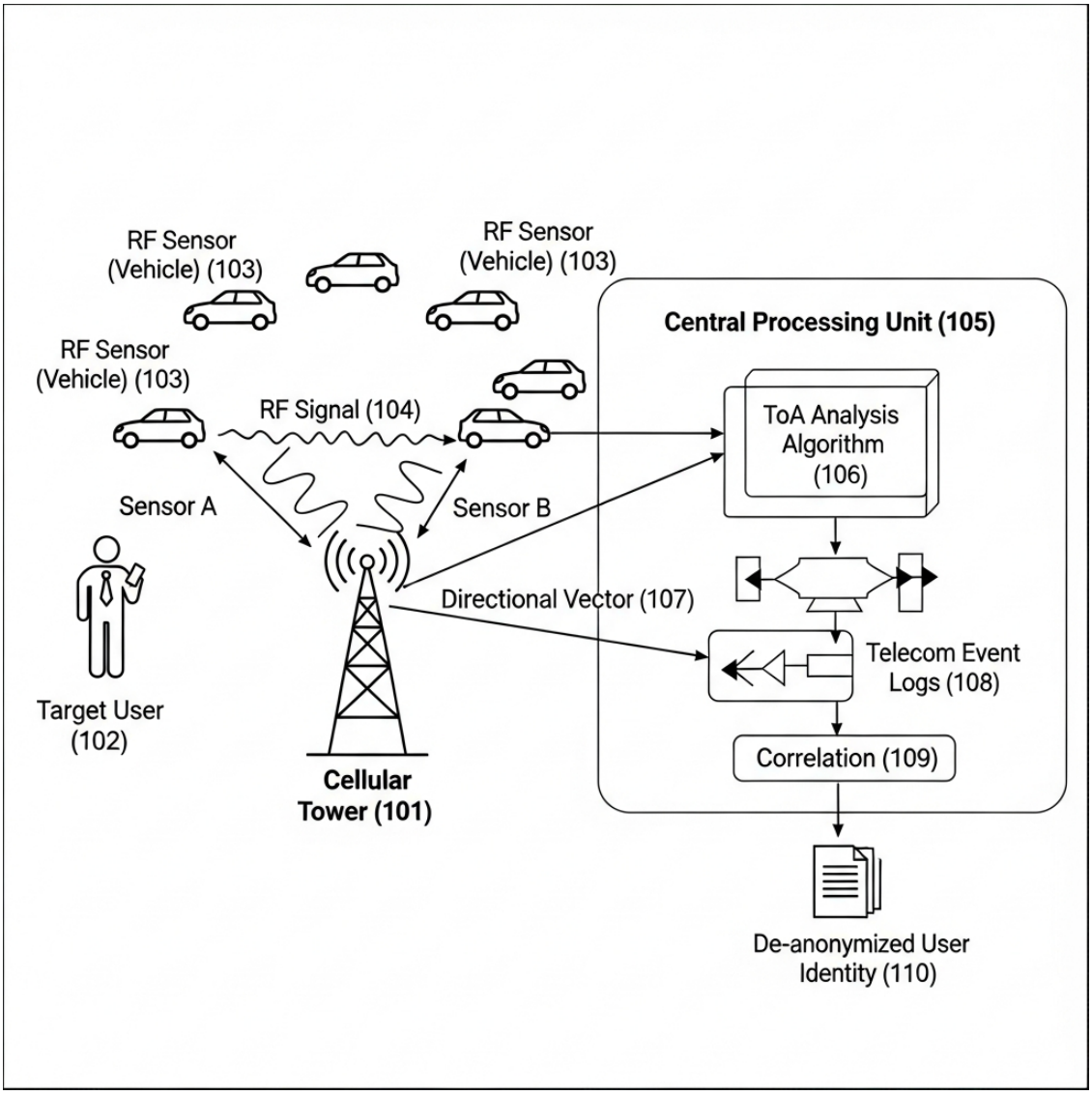


Figure 1: System diagram showing the overall architecture.

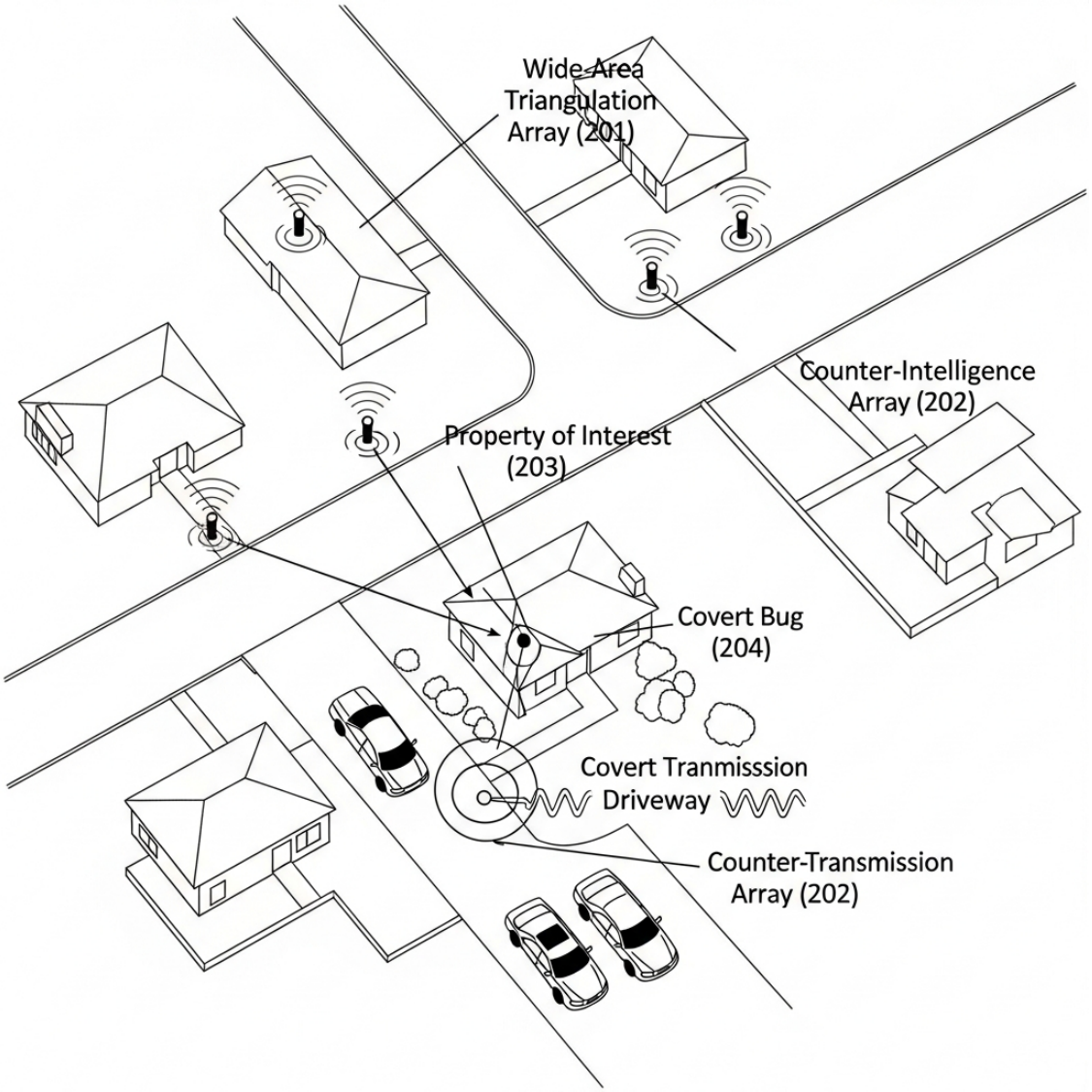


Figure 2: Conceptual diagram illustrating the dual-use deployment.